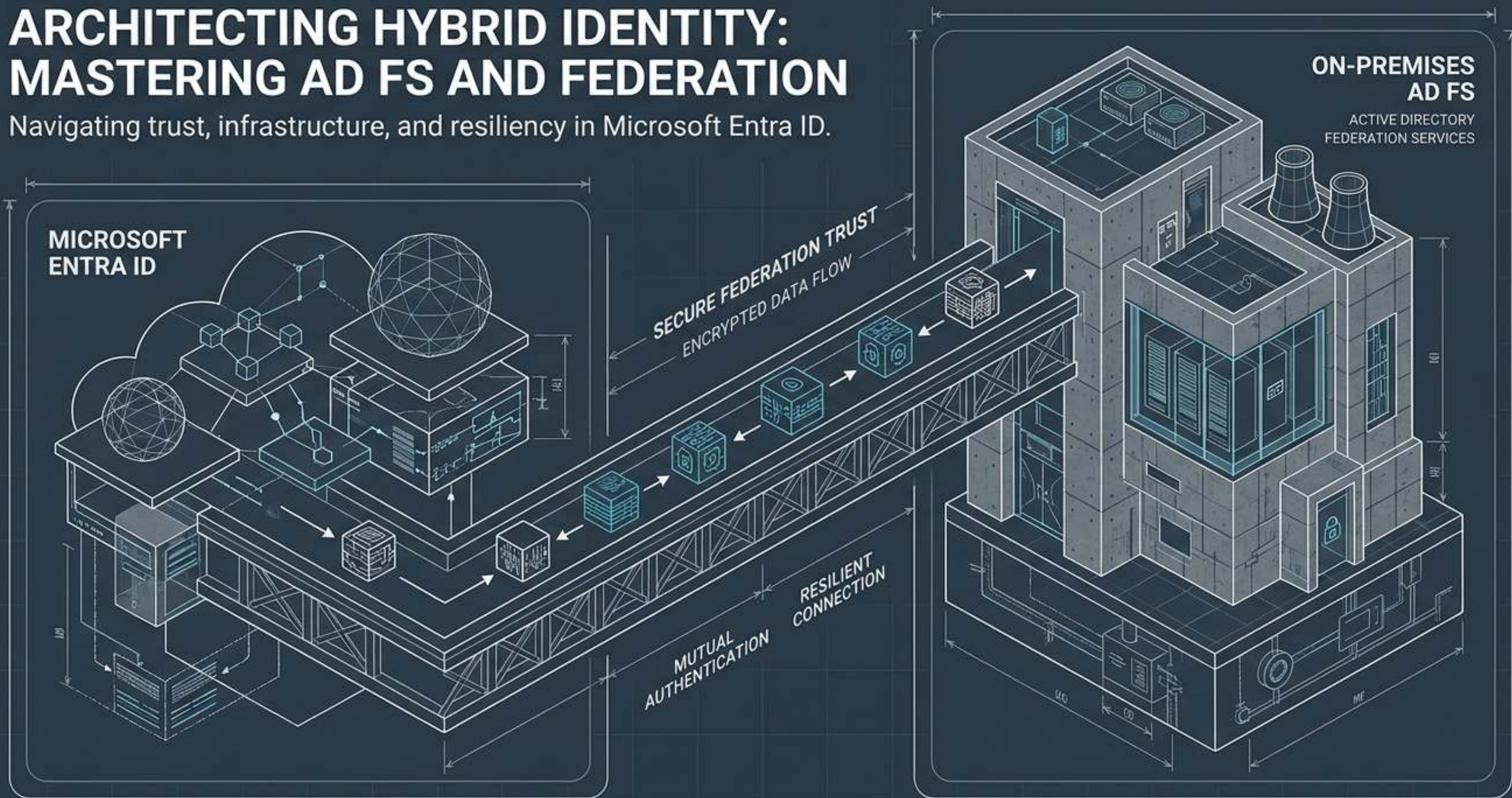


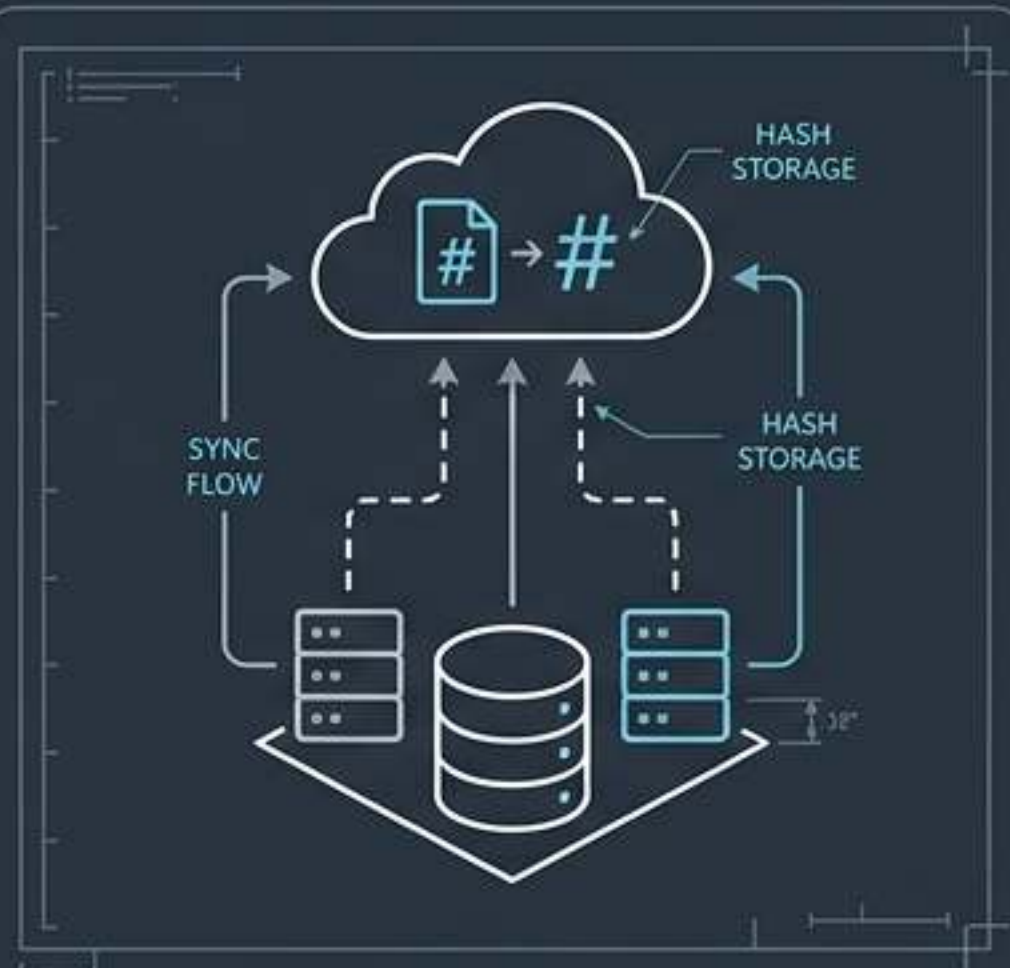
ARCHITECTING HYBRID IDENTITY: MASTERING AD FS AND FEDERATION

Navigating trust, infrastructure, and resiliency in Microsoft Entra ID.



THREE PRIMARY PILLARS SUPPORT HYBRID AUTHENTICATION IN ENTRA ID

CLOUD AUTH



PASSWORD HASH SYNCHRONIZATION (PHS)

Syncs a hash of on-prem passwords to Entra ID. Authentication happens purely in the cloud. Simplest infrastructure.

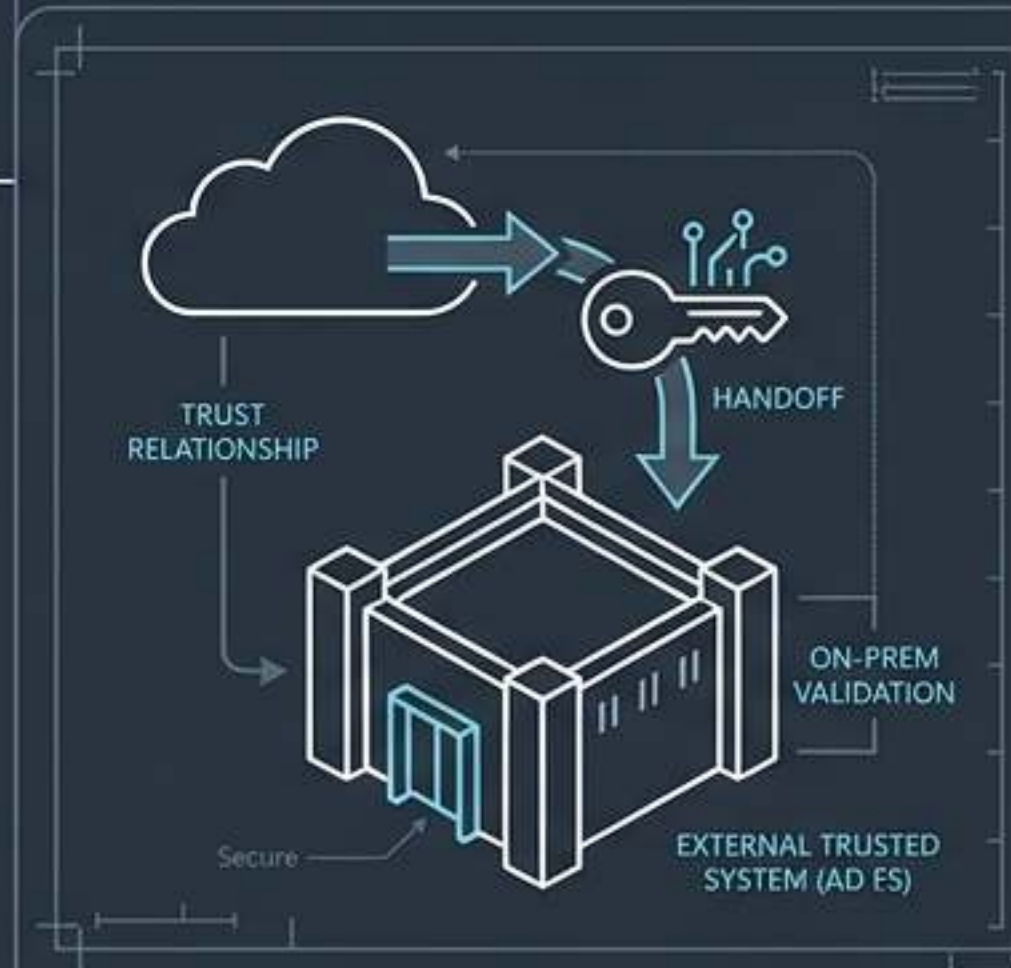
FEDERATED AUTH



PASS-THROUGH AUTHENTICATION (PTA)

Uses lightweight on-prem agents to validate passwords directly against AD DS. Authentication happens in the cloud after a secure on-prem verification exchange.

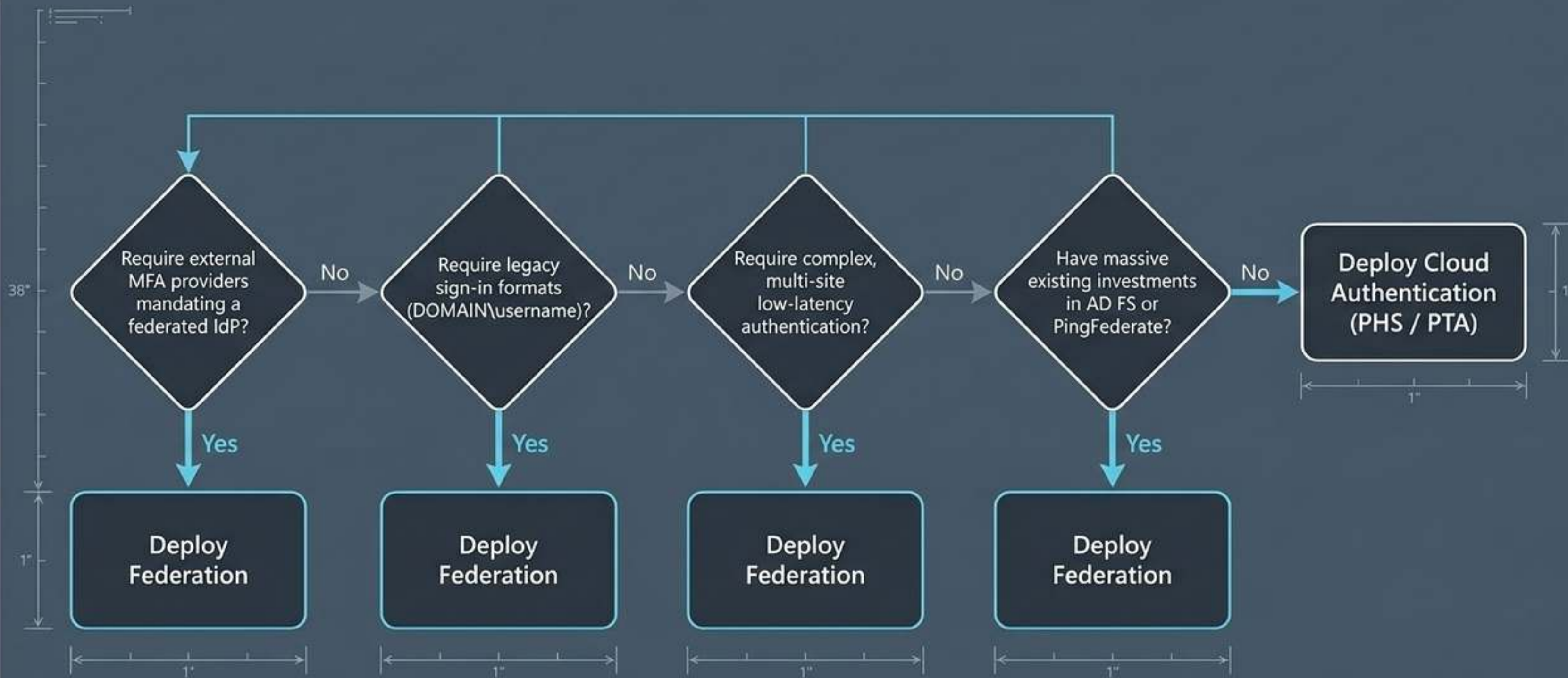
FEDERATED AUTH



FEDERATED AUTHENTICATION (AD FS)

Entra ID completely hands off the authentication process to an external trusted system. Validation happens entirely on-premises.

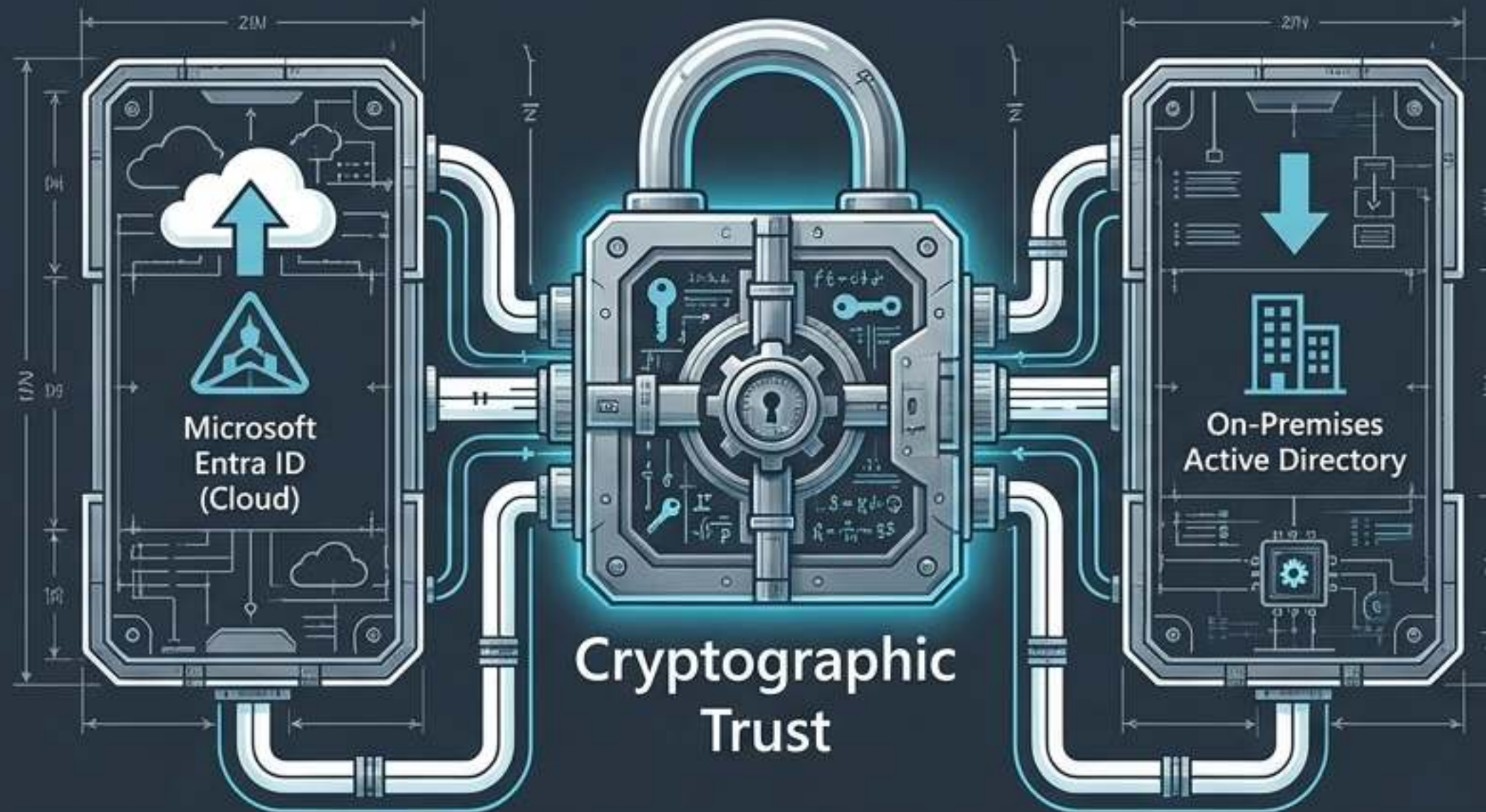
Edge cases and legacy investments mandate a federated architecture



The diagnostic matrix: Evaluating infrastructure, networking, and capability

Capability	Password Hash Sync (PHS)	Pass-Through Auth (PTA)	Federation (AD FS)
Where auth happens	Cloud	Cloud (via agent check)	On-premises
Server requirements	None	1+ authentication agents	2+ AD FS servers; 2+ WAP servers in DMZ
Network requirements	None beyond Connect	Outbound internet only	Inbound internet to WAP; inbound network access to AD FS; Load balancing
TLS/SSL Certificate	No	No	Yes
Account states supported	Disabled accounts (up to 30-min delay)	Immediately enforces disabled, locked, expired, sign-in hours	Immediately enforces all on-prem states
Sign-in types	UPN, Alternate ID, Certs	UPN, Alternate ID, Certs	UPN, sAMAccountName, Certs, Alternate ID

Federation establishes a cryptographic boundary of absolute trust



**Architectural Metaphor: Isolated Platforms
Bridged by Cryptographic Trust**



The Trust Definition

Federation is a cryptographic trust relationship established between domains.



The Handoff

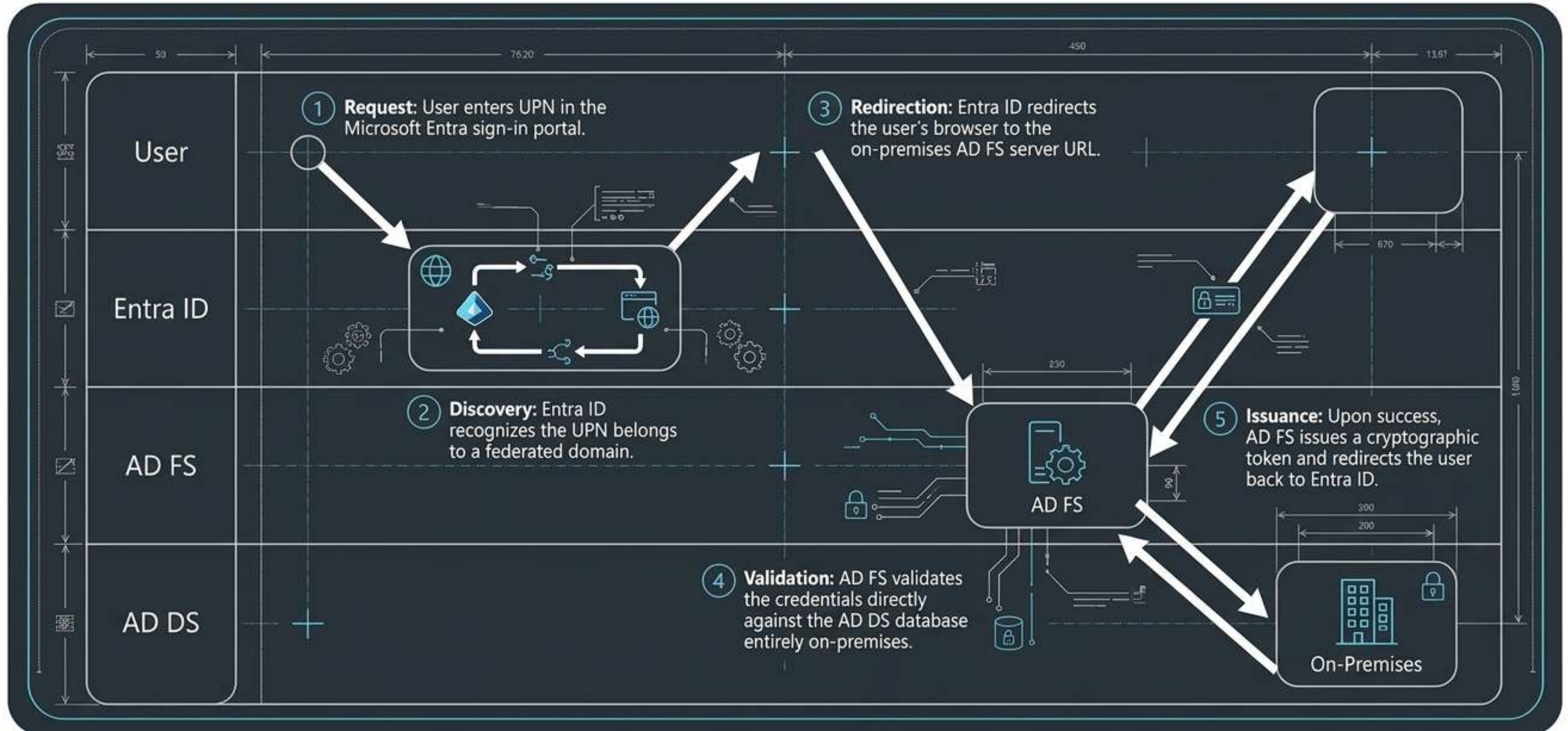
Entra ID steps out of the authorization path, completely handing off authentication to the trusted on-premises system.



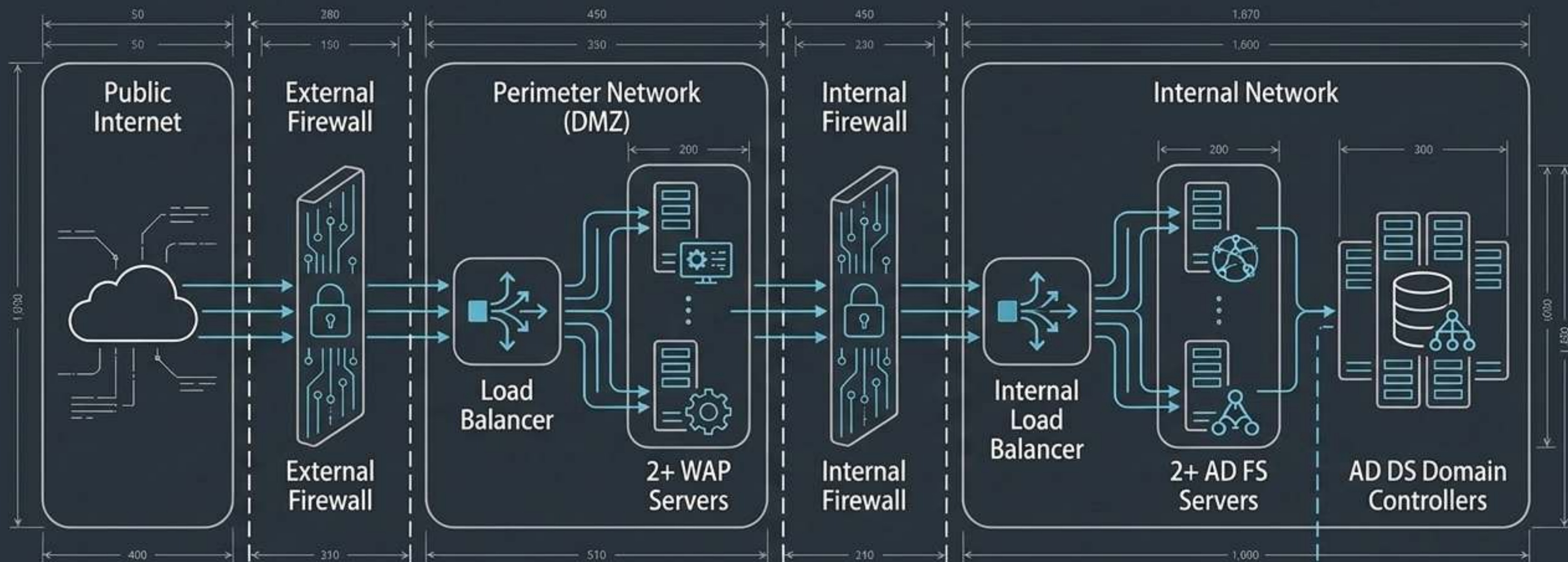
The Responsibility

Microsoft manages the cloud, but the ongoing operation, high availability, and security of the federated system becomes entirely the organization's responsibility.

The authentication handoff occurs entirely outside the cloud



Building the perimeter: The AD FS infrastructure blueprint

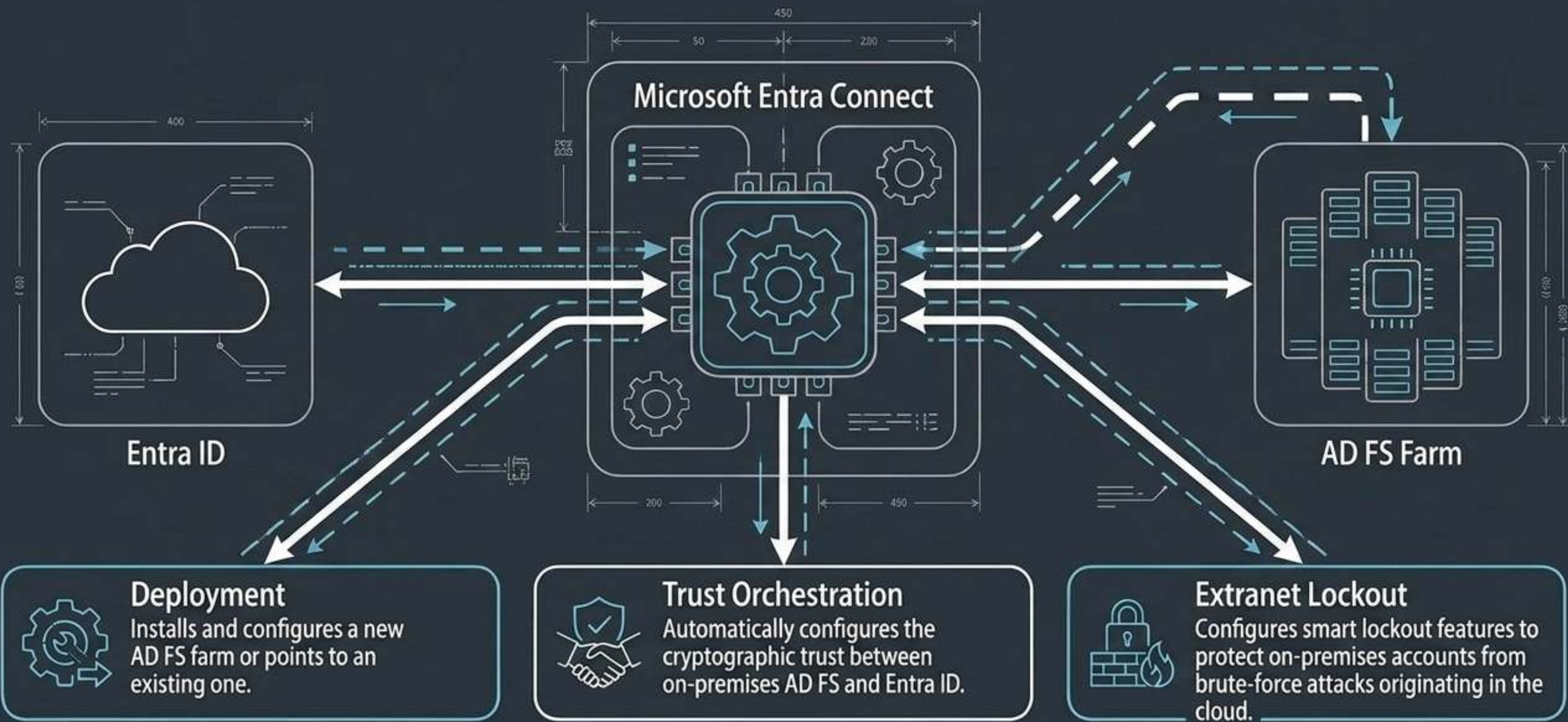


Requires inbound internet access to WAP servers.

Requires specific inbound network ports opened from WAP to AD FS.

Mandates public TLS/SSL certificates for the federation namespace.

Microsoft Entra Connect orchestrates the federation deployment



Day 2 Operations: Owning the authentication infrastructure



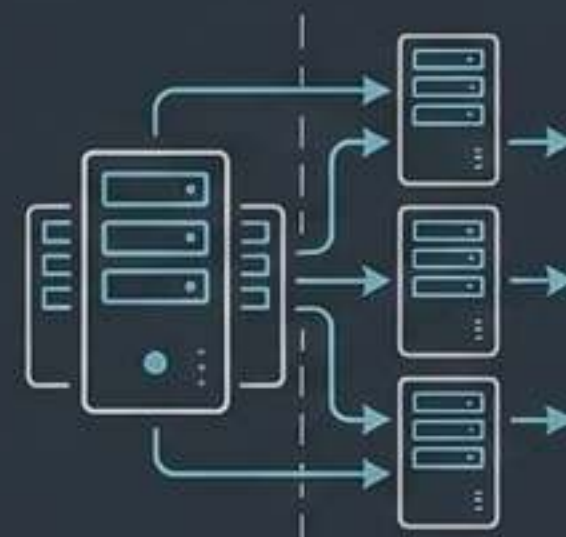
Certificate Management

Monitor and renew O365 and TLS/SSL federation certificates before expiration.



Repairing Trust

Actively monitor and repair the cryptographic trust relationship between AD FS and Azure/M365.



Scaling the Farm

Deploy and register additional AD FS or WAP servers to handle increased authentication loads.



Domain Federation

Configure and add new customized domains to be federated with the Entra ID instance.



Claim Rules

Modify or add AD FS claim rules corresponding to the organization's evolving access configurations.

The critical fail-safe: Password Hash Sync as a disaster recovery net



Always enable PHS

Even if AD FS is your primary method, PHS must run in the background to avoid total dependency on local infrastructure.

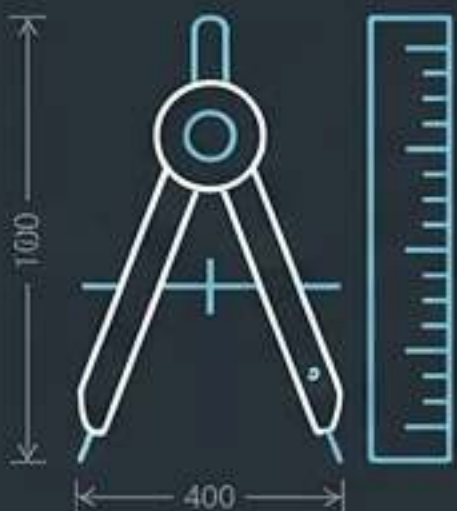
Business Continuity

If AD FS fails, admins can manually switch primary auth to PHS, restoring cloud access within hours.

Identity Protection

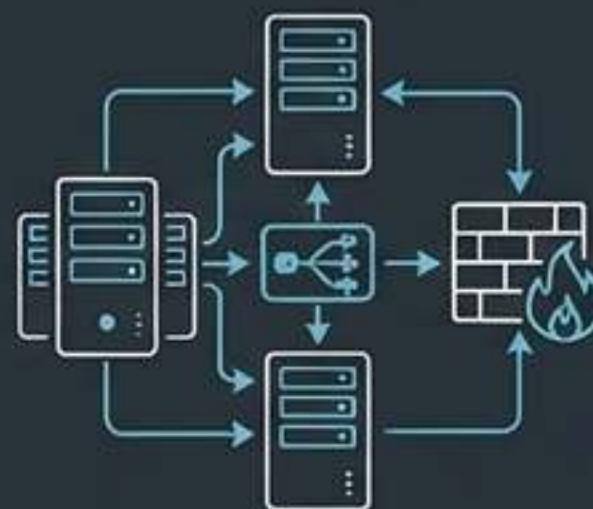
Enabling PHS is the only way to unlock Entra ID Protection's dark-web leaked-credential reports for hybrid accounts.

Executive Synthesis: Architecting resilient federated identity



The Mandate

Federation is powerful but should be reserved for strict edge cases like legacy sAMAccountName sign-ins and third-party MFA integrations.



The Blueprint

Requires rigorous on-premises infrastructure, mandating load-balanced AD FS servers internally and WAP servers in the DMZ.



The Operations

Federation shifts the operational burden to the enterprise, requiring diligent TLS certificate renewal and trust monitoring via Entra Connect.



The Fail-Safe

True Zero Trust resiliency requires redundancy. Password Hash Synchronization must be enabled as a mandatory safety net.